



3 Complex Analysis 2024

Tutorial 4

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Power series

Problem 16 Show that the series $\sum_{n=1}^{\infty} \frac{z^n}{n^2}$ converges absolutely for all $z \in \mathbb{C}$ with $|z| \leq 1$, and diverges for all $z \in \mathbb{C}$ with $|z| > 1$. In this case, the domain of convergence is a closed disc.

Solution.

By the Ratio Formula,

$$\frac{1}{R} = \lim_{n \rightarrow \infty} \frac{n^2}{(n+1)^2} = 1,$$

and the radius of convergence is $R = 1$. For $|z| \leq 1$ we have $\left| \frac{z^n}{n^2} \right| \leq \frac{1}{n^2}$, and the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent. Thus, by the Weierstrass M-Test the series $\sum_{n=1}^{\infty} \frac{z^n}{n^2}$ converges absolutely (and uniformly) in the closed disc $\{z \in \mathbb{C} : |z| \leq 1\}$.

Problem 17 Show that the series $\sum_{n=1}^{\infty} n z^n$ converges absolutely for all $z \in \mathbb{C}$ with $|z| < 1$, and diverges for all $z \in \mathbb{C}$ with $|z| \geq 1$. In this case, the domain of convergence is an open disc.

Solution.

By the Ratio Formula,

$$\frac{1}{R} = \lim_{n \rightarrow \infty} \frac{n+1}{n} = 1,$$

and the radius of convergence is $R = 1$. For $|z| \geq 1$ we have $|n z^n| \geq n$, and thus the terms do not tend to 0. It follows that the series diverges for all z with $|z| \geq 1$. The domain of convergence is the open disc $\{z \in \mathbb{C} : |z| < 1\}$.

Problem 18 Determine the radii of convergence of the following power series:

- | | |
|--|--|
| (a) $\sum_{n=0}^{\infty} (-1)^n (z-i)^n$ | (b) $\sum_{n=0}^{\infty} (-1)^n 2^n z^{2n+2}$ |
| (c) $\sum_{n=1}^{\infty} \left(\frac{n^2}{(2n+1)(n+2)} \right)^n z^n$, | (d) $\sum_{n=1}^{\infty} \frac{z^n}{\sqrt{n}}$, |
| (e) $\sum_{n=0}^{\infty} \frac{n!}{n^n} (z+2)^n$, | (f) $\sum_{n=0}^{\infty} 2^n z^{(2^n)}$. |

Solution.

- (a) This is in fact a geometric series and therefore $R = 1$. The Cauchy-Hadamard Formula and the Ratio Formula will, of course, give the same result:

$$\frac{1}{R} = \lim_{n \rightarrow \infty} |(-1)^n|^{\frac{1}{n}} = 1,$$

or

$$\frac{1}{R} = \lim_{n \rightarrow \infty} \frac{|(-1)^{n+1}|}{|(-1)^n|} = 1.$$

- (b) Here, I think, the easiest way is to use the Ratio Test for $\sum_{n=0}^{\infty} |(-1)^n 2^n z^{2n+2}|$, namely,

$$\lim_{n \rightarrow \infty} \frac{2^{n+1} |z|^{2(n+1)+2}}{2^n |z|^{2n+2}} = 2|z|^2 < 1 \quad \Longleftrightarrow \quad |z| < \frac{1}{\sqrt{2}}.$$

The radius of convergence is $R = \frac{1}{\sqrt{2}}$.

- (c) By the Cauchy-Hadamard Formula,

$$\frac{1}{R} = \limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \frac{n^2}{(2n+1)(n+2)} = \frac{1}{2},$$

and $R = 2$.

- (d) The Ratio Formula gives

$$\frac{1}{R} = \lim_{n \rightarrow \infty} \frac{\sqrt{n}}{\sqrt{n+1}} = 1,$$

and $R = 1$.

- (e) By the Ratio Formula,

$$\frac{1}{R} = \lim_{n \rightarrow \infty} \frac{(n+1)!}{(n+1)^{n+1}} \frac{n^n}{n!} = \lim_{n \rightarrow \infty} \frac{n^n}{(n+1)^n} = \lim_{n \rightarrow \infty} \frac{1}{\left(\frac{n+1}{n}\right)^n} = \frac{1}{\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n} = \frac{1}{e},$$

and $R = e$.

- (f) Notice that in this example we have

$$a_m = \begin{cases} m, & \text{if } m = 2^n \text{ with } n = 0, 1, 2, \dots, \\ 0, & \text{otherwise.} \end{cases}$$

By the Cauchy-Hadamard Formula,

$$\frac{1}{R} = \limsup_{m \rightarrow \infty} |a_m|^{\frac{1}{m}} = \lim_{m \rightarrow \infty} m^{\frac{1}{m}} = 1.$$

Problem 19 Use the geometric series to determine the power series expansion of the form $\sum_{n=0}^{\infty} a_n z^n$ for the following functions:

$$(a) \frac{1}{2z+5}, \quad (b) \frac{1}{1+z^4}, \quad (c) \frac{1+iz}{1-iz}, \quad (d) \frac{1}{(z+1)(z+2)}.$$

For each series, give its domain of convergence.

Solution.

Recall the geometric series $\sum_{n=0}^{\infty} w^n = \frac{1}{1-w}$; it converges if and only if $|w| < 1$.

(a) We have

$$\frac{1}{2z+5} = \frac{1}{5} \cdot \frac{1}{1+\frac{2}{5}z} = \frac{1}{5} \sum_{n=0}^{\infty} \left(-\frac{2}{5}z\right)^n = \sum_{n=0}^{\infty} \frac{(-1)^n 2^n}{5^{n+1}} z^n.$$

It converges if and only if $\left|\frac{2}{5}z\right| < 1$, i.e., the domain of convergence is $\{z \in \mathbb{C} : |z| < \frac{5}{2}\}$.

(b) We have

$$\frac{1}{1+z^4} = \sum_{n=0}^{\infty} (-z^4)^n = \sum_{n=0}^{\infty} (-1)^n z^{4n}.$$

It converges if and only if $|z^4| < 1$. The domain of convergence is $\{z \in \mathbb{C} : |z| < 1\}$.

(c) We have

$$\begin{aligned} \frac{1+iz}{1-iz} &= (1+iz) \frac{1}{1-iz} = (1+iz) \sum_{n=0}^{\infty} (iz)^n = \sum_{n=0}^{\infty} (iz)^n + \sum_{n=0}^{\infty} (iz)^{n+1} \\ &= 1 + \sum_{n=1}^{\infty} 2i^n z^n. \end{aligned}$$

It converges if and only if $|iz| < 1$. The domain of convergence is $\{z \in \mathbb{C} : |z| < 1\}$.

(d) We have

$$\begin{aligned} \frac{1}{(z+1)(z+2)} &= \frac{1}{z+1} - \frac{1}{z+2} = \sum_{n=0}^{\infty} (-z)^n - \frac{1}{2} \sum_{n=0}^{\infty} \left(-\frac{z}{2}\right)^n \\ &= \sum_{n=0}^{\infty} (-1)^n \left(1 - \frac{1}{2^{n+1}}\right) z^n. \end{aligned}$$

The geometric series for $\frac{1}{z+1}$ converges if and only if $|z| < 1$. The geometric series for $\frac{1}{z+2}$ converges if and only if $\left|\frac{z}{2}\right| < 1$. Thus, the domain of convergence is $\{z \in \mathbb{C} : |z| < 1\}$.